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Trajectory Optimization Strategy for Networked UAVs Based on Deep Learning

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ABSTRACT

The rapid advancement of networked UAV technology has broadened its applications in logistics, disaster relief, agriculture, and military. However, precise trajectory planning in complex dynamic environments remains a key challenge. Traditional methods suffer from low computational efficiency and poor real-time response, limiting practical use. This paper proposes a deep learning-based trajectory optimization strategy for networked UAVs, integrating reinforcement learning and deep neural networks to autonomously learn environmental information and collaborative patterns. Simulation experiments in realistic scenarios validate its effectiveness: the method enhances trajectory planning accuracy and robustness while reducing computational complexity and improving real-time response. This research provides new theoretical and technical foundations for promoting networked UAV applications.

KEYWORDS

Deep Learning, Networked UAVs, Trajectory Optimization, Strategy

INTRODUCTION

In recent years, the vigorous development of unmanned aerial vehicle (UAV) technology has brought broad application prospects (Fei et al., 2023; Rinaldi & Primatesta, 2024). UAVs have become indispensable tools in modern society, spanning scenarios such as military reconnaissance, logistics distribution, environmental monitoring, and post-disaster relief (Amrallah et al., 2023). Particularly driven by networked technologies, the advantages of collaborative operations in UAV swarms have become increasingly prominent (Luo et al., 2022; Zhang et al., 2022). Multi-vehicle cooperation and information sharing have significantly improved the efficiency and reliability of task execution (Wang & Gursoy, 2021). Networked UAV collaboration brings new challenges to trajectory optimization. It is more complex than single-UAV scenarios. In a swarm, UAVs must interact constantly. They share real-time data like positions and tasks. But communication delays or interruptions can break this interaction. This makes coordinated trajectory planning harder. Also, multiple UAVs pursue common goals while handling their own tasks. Their paths must avoid collisions. They also need to share limited resources like airspace and energy. Balancing these needs adds difficulty to trajectory design. Dynamic environments make it worse. Sudden obstacles or weather changes require all UAVs to adjust paths quickly. Without seamless coordination, the whole mission may fail. In urban logistics, UAV swarms can collaboratively plan routes to avoid buildings and other aircraft. This ensures timely delivery of medical supplies to remote communities (Li et al., 2022). For forest fire monitoring, optimized trajectories allow swarms to cover larger areas. They can quickly adjust

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paths when flames spread, ensuring continuous data collection (Xu et al., 2022). In post-earthquake rescue, UAVs with optimized trajectories can navigate through collapsed structures. They coordinate to locate survivors and send real-time images to rescue teams (Adam et al., 2024; Demiane et al., 2020). Nevertheless, the trajectory optimization problem of UAVs in complex environments still faces severe challenges, such as rapid dynamic environmental changes, significant communication delays, and diverse constraints on UAV endurance and safety. These issues urgently require effective trajectory optimization strategies to enable UAVs to better adapt to real-world application needs (Han et al., 2025; Xu et al., 2020).

Traditional UAV trajectory optimization methods mainly rely on heuristic algorithms, such as genetic algorithms and rapidly exploring random trees (RRT). Although these methods possess certain optimization capabilities, they still fall short in real-time performance and adaptability to complex scenarios (Yu et al., 2022). Meanwhile, deep learning technology has achieved remarkable results in fields such as image recognition and natural language processing (Wu et al., 2021). Its powerful adaptive learning and environmental perception capabilities provide new possibilities for trajectory optimization (Hua et al., 2021; Yang & Liu, 2024). Against this backdrop, this paper introduces deep learning into the field of trajectory optimization for networked UAVs and proposes an optimization strategy that integrates reinforcement learning and deep neural networks. The goal is to significantly improve the accuracy and real-time performance of trajectory planning. Through systematic theoretical analysis and comprehensive simulation verification, this paper explores a technical pathway to effectively enhance the trajectory optimization performance of UAVs, thereby better meeting the efficient and intelligent requirements of networked UAVs in practical applications.

LITERATURE REVIEW

In recent years, with the rapid development of deep learning technology, some scholars have begun to explore the integration of deep learning and UAV trajectory optimization, achieving encouraging progress. Zeng et al. (2020) studied UAV-enabled wireless networks where a UAV acts as an aerial base station, jointly optimizing user scheduling, trajectory, transmit power, and bandwidth allocation to meet diverse user QoE requirements under resource and energy constraints. Their iterative algorithm achieved 12.5% higher energy efficiency than throughput-maximizing baselines. Han et al. (2021) further expanded on communication-related trajectory optimization, focusing on UAV-aided backscatter communications; they analyzed performance bottlenecks and optimized trajectories to balance communication reliability and energy efficiency, providing new insights for trajectory design in low-power communication scenarios. Deng et al. (2022) extended this direction to UAV-enabled mobile-edge computing, proposing joint optimization of model decision, resource allocation, and trajectory for AI applications, which enriched the integration of trajectory optimization with edge computing tasks.

The action mask mechanism, target capture mechanism, and collision avoidance mechanism proposed in the AM-MAPPO algorithm by Liu et al. (2024) have played positive roles in improving algorithm effectiveness, target capture capability, and UAV safety, respectively, providing a potential and effective reference direction for practical project applications. Pan et al. (2021) combined deep learning with genetic algorithms, using genetic algorithms to train deep learning models for multi-UAV data collection path planning, which enhanced the efficiency of path optimization in collaborative data collection scenarios and expanded the application of hybrid algorithms in trajectory optimization.

Hu et al. (2022) explored trajectory planning for cellular-connected UAVs in communication-assisted radar sensing scenarios. They designed a BS-UAV bistatic SAR platform and proposed a new trajectory planning algorithm based on successive convex approximation and block coordinate descent, aiming to minimize UAV propulsion energy while ensuring required sensing resolutions. Simulation results showed that the algorithm outperformed distance-minimizing alternatives in energy efficiency, achieving up to 55% energy savings. In smart city scenarios, Zema et al. (2022) focused on 3D

trajectory optimization for multimission UAVs, addressing the challenge of coordinating multiple tasks (e.g., surveillance, delivery) in urban airspace and providing a multiobjective optimization framework for complex urban environments.

Partsinevelos et al. (2020) proposed a novel UAV-assisted positioning system for GNSS-denied environments, where UAS, with its higher elevation ensuring clear-sky visibility to GNSS satellites, transfers its on-board GNSS accuracy to the ground. They described the system's hardware/software architecture, coordinate system integration, and conducted error budget analysis to evaluate its potential.

Ji et al. (2020) focused on multi-UAV assisted wireless networks, aiming to maximize the minimum throughput of UAV-served users by jointly optimizing cache placement, trajectory, and transmission power under limited storage. They proposed an alternating iterative algorithm to solve the mixed-integer non-convex problem, with numerical results showing significant throughput enhancement compared to benchmarks lacking trajectory design and power control. Khoufi et al. (2021) complemented this by studying UAV trajectory optimization for data pick-up and delivery with time windows, emphasizing the importance of time constraints in multi-UAV collaborative tasks and providing a practical reference for time-sensitive logistics scenarios. Singh et al. (2022) deployed a semi-autonomous UAV swarm with a master-slave architecture (three small UAVs) for high-resolution linear photogrammetric surveys, specifically river bed mapping of a 13.3 ha area of Solani River. Using 5 GCPs for geo-referencing and Pix4DMapper for data processing, the swarm achieved 1.24 cm GSD and 0.023 m RMS error in geospatial results, significantly reducing survey time and cost.

In practical application scenarios, Pyra and Żołądek (2025) applied UAVs and photogrammetric techniques in a basalt quarry in Lower Silesia to improve blast design accuracy and mitigate flyrock. Using a DJI Mavic 3 Enterprise UAV and Strayos software to create 3D models with centimeter-level accuracy, they identified improper burden zones and made real-time blasthole design corrections. This methodology reduced flyrock range by an average of 42% near sensitive areas, even with a ten-fold reduction in images used, demonstrating enhanced safety and operational benefits in the actual quarry project. Munawar et al. (2021) integrated GIS, remote sensing and UAV technologies to build a holistic model for bushfire assessment and mitigation in their study on the 2020 Victorian bushfires. In practical implementation, using VIIRS and MODIS fire products, they found 1500m aggregates best for estimating burnt areas, identified fire hotspots concentrating in eastern Victoria and spreading north to New South Wales, with R^2 values of 0.91-0.99 indicating high method fitness and z-values of 0.03-2.9 showing statistical significance. They optimized UAV paths with five algorithms, among which PSO performed best in runtime and cost, with an average improvement of ~2% over the greedy algorithm, 1.2% over the intra-route algorithm, and cost reductions of 1.5% vs inter-route and 1.2% vs intra-route, providing an efficient and feasible solution for local disaster management authorities to assess and respond to bushfires.

In environmental monitoring, Jeyaraj et al. (2022) conducted a study on Malgund Beach along India's Ratnagiri coast, using UAVs to monitor beach sediment changes. A UAV with a 4k camera captured images within a 6km range, and the collected data was used to generate a high-resolution Digital Elevation Model (DEM), which was validated with Ground Control Points (GCPs). The validated dataset showed a good correlation coefficient ($r = 0.86$). Through the obtained cross-sectional profiles, coastal erosion was successfully identified and quantified, demonstrating the effectiveness of UAVs in monitoring short-term changes in beach morphology and sediments. Liu et al. (2022) supplemented this field by studying UAV trajectory optimization for time-constrained data collection in environmental monitoring systems, highlighting the role of time efficiency in large-scale environmental surveys.

Although existing research provides a rich methodological foundation and empirical accumulation for trajectory optimization, there is still a lack of systematic and in-depth exploration and solutions for real-time trajectory optimization and communication constraints in the context of multi-UAV collaboration for networked UAVs.

MATERIALS AND METHODS

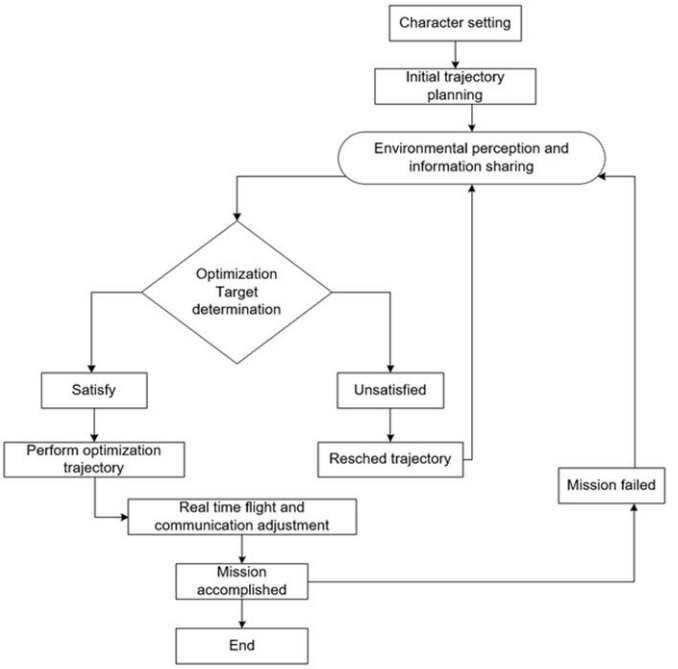
Modeling of Trajectory Optimization Problem for Networked UAVs

The essence of the UAV trajectory optimization problem is to enable UAVs to efficiently and safely complete tasks under given mission objectives, which involves multiple constraints such as path length, flight safety, energy consumption, and communication quality. In practical scenarios of networked UAVs, communication and collaborative interaction between each UAV are key factors determining the quality of task completion; in other words, the single-UAV optimization problem is transformed into a complex optimization problem of multi-UAV collaboration. To accurately characterize this feature, this paper establishes a mathematical model that comprehensively considers flight constraints, environmental impacts, and network communication characteristics, abstracting UAV trajectory optimization as a typical multi-objective optimization problem (Liu et al., 2023).

Specifically, the model first defines a set of state variables, including the UAV's spatial position, velocity, energy consumption level, and current communication status. Meanwhile, a multi-objective function is constructed with the objectives of minimizing trajectory length, minimizing energy consumption, and minimizing network delay (Chen et al., 2023). At the same time, the constraints involve kinematic constraints, dynamic constraints, obstacle avoidance constraints, and communication distance constraints between UAVs. It is worth noting that this paper particularly emphasizes the importance of communication constraints, which are critical for collaborative optimization of networked UAVs. The model addresses the multi-objective optimization problem in the following manner: a weighted summation method is used to comprehensively evaluate multiple objectives to obtain a reasonable trade-off solution; a constraint determination mechanism is introduced to ensure that the solution process remains within the effective solution space (Wang et al., 2024).

To clearly describe the logical framework of the trajectory optimization model for networked UAVs, this paper provides a basic framework diagram of the problem (Figure 1). As shown in the framework diagram, the trajectory optimization process is a closed-loop feedback structure. Through the state detection and real-time feedback mechanism, UAVs can autonomously adjust their flight trajectories according to changes in the dynamic environment and network status. In addition, the establishment of this model provides a clear foundation of optimization objectives and constraint conditions for the subsequent application of deep learning methods.

Figure 1. Basic Framework of Trajectory Optimization for Networked UAVs



Theoretical Foundation of UAV Trajectory Optimization Based on Deep Learning

With the continuous maturity of deep learning technology, it has begun to demonstrate increasingly strong advantages and potential in the field of trajectory optimization (Liu et al., 2023). Especially after breakthroughs in areas such as Reinforcement Learning (RL) and Deep Neural Networks (DNN), scholars have started to apply these technologies to UAV trajectory optimization (Liu et al., 2024). The greatest advantage of reinforcement learning lies in its ability to autonomously learn and optimize strategies through environmental feedback without explicit modeling; deep neural networks excel at processing high-dimensional and non-linear data, effectively improving optimization accuracy and generalization capabilities through feature extraction (Chen et al., 2023). Against this backdrop, many reinforcement learning models have been attempted for UAV trajectory optimization in recent years (Zhang et al., 2022).

The Deep Q-Network (DQN) can effectively solve problems in discrete decision spaces but performs poorly in continuous action spaces; while the Deep Deterministic Policy Gradient (DDPG) algorithm overcomes this limitation, its training efficiency significantly decreases in highly complex environments (Zhang et al., 2022). In addition, temporal models such as neural network structures like Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU), due to their good ability to capture temporal features, have gradually been applied to the field of trajectory prediction and optimization, demonstrating promising potential and effects (Li et al., 2024).

To better illustrate the differences and characteristics of different deep learning methods, Table 1 provides a performance comparison of current mainstream deep learning algorithms in UAV trajectory optimization. As shown in Table 1, different methods have their own advantages and disadvantages in terms of generalization performance, computational efficiency, and real-time responsiveness, which provides an important theoretical basis for the selection and improvement of deep learning models in this paper.

Table 1. Performance Comparison of Different Deep Learning Methods in UAV Trajectory Optimization

Method	Generalization Error (%)	Computational Efficiency (ms)	Response Delay (ms)
DQN	12.4	150	120
DDPG	9.8	200	110
LSTM	7.2	180	100

Analysis of Existing Problems in UAV Trajectory Optimization Models

Although deep learning-based UAV trajectory optimization models have made significant progress in both theoretical and practical aspects in recent years, a series of notable issues persist in real-world applications (Chen et al., 2023). First, most existing deep learning trajectory optimization models are overly dependent on the quality of training datasets. When there is a significant discrepancy between the real environment and the distribution of the training dataset, the generalization performance of the models significantly declines, exhibiting obvious instability (Wang et al., 2022). Additionally, model training typically requires long computation times and high computational resource demands, making it difficult to guarantee real-time response capabilities and restricting the expansion of practical application scenarios (Zhang et al., 2024).

Second, from the perspective of algorithm structures, although models such as DQN and DDPG have achieved certain successes, they have not fully leveraged their advantages when facing multi-UAV collaboration and communication issues (Li et al., 2023). Especially in networked UAV environments, communication delays and network outages occur frequently, and these communication problems have not been deeply integrated into model considerations, leading to often unsatisfactory actual operation effects (Wang et al., 2022). Furthermore, most current trajectory optimization models do not fully account for energy consumption constraints and endurance issues. In practical applications, UAVs frequently fail to complete tasks due to energy consumption exceeding expectations, which further limits the effective implementation of deep learning methods.

Table 2 presents the performance metrics of current mainstream optimization models, such as generalization error, computational complexity, response delay, and energy efficiency. As shown in Table 2, while existing models have their respective advantages, they commonly suffer from high computational complexity, large response delays, and insufficient energy consumption management. There is an urgent need to develop an optimization model or strategy that can efficiently handle multi-UAV collaboration issues while balancing computational efficiency and practical energy consumption management to truly meet the diverse needs of real-world applications.

Table 2. Performance Index Data Comparison of Existing UAV Trajectory Optimization Models

Model Type	Generalization Error (%)	Computational Complexity (Operation Count×10^3)	Response Delay (ms)	Energy Management Efficiency (%)
Traditional Method	15.5	250	200	60
DQN Method	12.4	220	120	70
DDPG Method	9.8	300	110	68

RESULTS

Proposed Deep Learning Networked Trajectory Optimization Model

Aiming at the problems of insufficient generalization and low computational efficiency in current UAV trajectory optimization models, this paper proposes a deep learning networked trajectory optimization model that fuses reinforcement learning and temporal neural networks. Unlike traditional reinforcement learning models, this study innovatively combines a convolutional neural network (CNN) with a long short-term memory network (LSTM) to construct a spatio-temporal fusion optimization framework. Specifically, the model first uses a CNN to extract feature information about the UAV's surrounding environment and spatial position, and then leverages an LSTM network to learn dynamic interaction patterns in continuous time series. This design not only improves the model's adaptability to complex dynamic environments but also effectively reduces computational complexity.

The fusion of CNN and LSTM in this model is implemented through a detailed feature-level integration process. First, the 3-layer CNN processes raw environmental inputs, which include 2D spatial data like obstacle coordinates, building outlines, and terrain elevation maps. Each convolutional layer applies filters to generate feature maps: the first layer outputs edge and texture features (e.g., small obstacle edges), the second layer aggregates these into local spatial patterns (e.g., clusters of obstacles), and the third layer produces high-level global scene features. These 2D feature maps are then flattened into a 1D feature vector of length 256, capturing all spatial characteristics. Next, this spatial feature vector is concatenated with temporal data streams at the input layer of the LSTM. The temporal data includes the UAV's past 10 timesteps of position coordinates, velocity vectors, and energy consumption rates. This concatenation creates a combined input vector that merges spatial and temporal features into a single tensor. The fused input tensor for LSTM is defined as Equation 1.

$$X_{fused} = Concat\left(X_{CNN}, X_{LSTM-temporal}\right) \tag{1}$$

Here, $X_{CNN} \in R^{256}$ stands for the 1D spatial feature vector extracted by the CNN, which encapsulates all spatial characteristics of the environment. $X_{LSTM\text{-}temporal} \in R^{3\times10}$ represents the temporal data, including the UAV's position coordinates, velocity vectors, and energy consumption rates recorded over the past 10 timesteps. The $Concat(\cdot)$operation combines these two components into a single tensor, resulting in $X_{fused} \in R^{286}$ that serves as the input for the LSTM network. This fused tensor enables the LSTM to simultaneously process both spatial context and temporal dynamics in trajectory optimization.

The LSTM network then processes this fused tensor sequentially. It updates its hidden state at each timestep by leveraging both the new spatial features (from CNN) and the accumulated temporal patterns (from historical data). This ensures that every decision in the LSTM considers both the current spatial context and how the UAV's trajectory has evolved over time. Unlike decision-level fusion, where CNN and LSTM would generate separate outputs before combining, this feature-level approach allows spatial and temporal information to interact throughout the LSTM's computation. This tight integration helps the model better capture complex relationships, such as how a sudden obstacle (spatial feature) should alter the trajectory trend (temporal pattern) over the next 5–8 timesteps.

To further enhance the robustness of communication between UAVs, the model also incorporates an adaptive communication control mechanism. This mechanism can real-time evaluate the network communication status and dynamically adjust the communication frequency between UAVs to avoid issues such as communication congestion or network disconnection. The details of the model architecture are clearly presented in Figure 2, through which the implementation path and specific process of the entire optimization strategy can be intuitively observed. The specific settings and values of the experimental parameters are shown in Table 3, which indicates that the main parameters of the model include key values such as the number of network layers, learning rate, and communication

frequency adjustment thresholds. By repeatedly tuning these parameters, the optimal parameter configuration was finally determined to ensure the model achieves optimal performance.

Figure 2. Framework Diagram of the Proposed Deep Learning Networked Trajectory Optimization Model

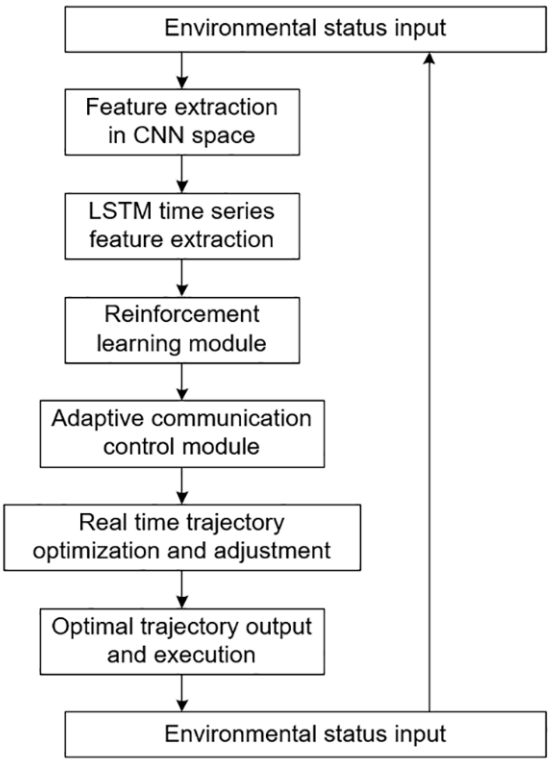


Table 3. Parameter Setting Data of the Proposed Model

Parameter Name	Parameter Value
Number of CNN Layers	3 layers
Number of LSTM Hidden Units	128 units
Learning Rate	0.001
Communication Frequency Threshold	20 ms
Batch Size	64

The 3-layer CNN is chosen for a reason. It handles spatial features like buildings and obstacles in urban environments. Urban scenes have medium complexity. Three layers work well here. The first layer catches small details, like streetlights. The second layer groups local features, like building clusters. The third layer forms a global view of the scene. This setup balances feature extraction and computing speed. It avoids overfitting that more layers might cause. LSTM deals with time-related changes, such as speed and obstacle movement. In multi-UAV tasks, we need to check 5–10 past time

steps to predict paths. 128 units store enough past information for short-term patterns. They also prevent extra computing that would slow down responses, which is key for time-sensitive tasks like emergency deliveries. The learning rate of 0.001 is based on tests. It affects how fast and stably the model learns. In dynamic cities, UAVs must adapt quickly to sudden changes, like new obstacles. Higher rates (e.g., 0.01) made training unstable. Lower rates (e.g., 0.0001) made learning too slow. 0.001 lets the model update fast while staying stable. The 20 ms communication threshold fits urban conditions. In dense cities, UAV communication delays usually range from 10 to 30 ms because buildings block signals. 20 ms triggers adjustments. If delay goes over 20 ms, the model reduces communication to avoid data loss. If delay is lower, it increases communication to share info on time. This balances reliability and energy use. The batch size of 64 matches the dataset and UAV hardware. The dataset has over 10,000 trajectory samples from simulated cities. 64 samples per batch help the model learn patterns from different cases, improving generalization. It also fits the memory limits of UAV processors, letting training and calculation run on the UAV without external help.

Experimental Design and Simulation Environment Setup

To comprehensively evaluate the practical performance of the proposed trajectory optimization model, this study designs a series of simulation experiments to verify the model's performance in complex dynamic environments. The experiments are conducted in a MATLAB environment, simulating real-world urban delivery scenarios while considering various practical challenges such as dynamic obstacles, communication delays, and network interruptions that UAVs may encounter (Ramanan et al., 2025).

(1) Experimental Scene Setup

The simulation constructs a delivery task scenario involving multiple UAVs, where each UAV is assigned a specific delivery task with clear start and end points. Meanwhile, a series of dynamic obstacles are randomly generated in the simulated environment, such as sudden building obstructions, adverse weather zones, or other flying objects, to assess the model's real-time obstacle avoidance capabilities. Additionally, the experiment simulates communication delays and signal interference in real network environments, making the simulation more aligned with actual application scenarios.

(2) Experimental Procedures

Model Training: The proposed deep learning model is first trained using a simulated dataset. The training process optimizes model parameters through iterative adjustments to enhance its adaptability to dynamic environments.

Real-Time Trajectory Optimization: After training, the model is deployed in the simulation environment to perform real-time trajectory optimization. Key indicators such as actual flight trajectories, response delays, and the number of obstacle avoidance actions for each UAV are recorded.

Visualization and Analysis: Figure 3 illustrates the simulated experimental environment and UAV trajectories, clearly demonstrating the real-time adjustment of UAV paths. This visualization aids in intuitively evaluating the model's performance in dynamic obstacle avoidance and trajectory planning.

(3) Simulation Environment Characteristics

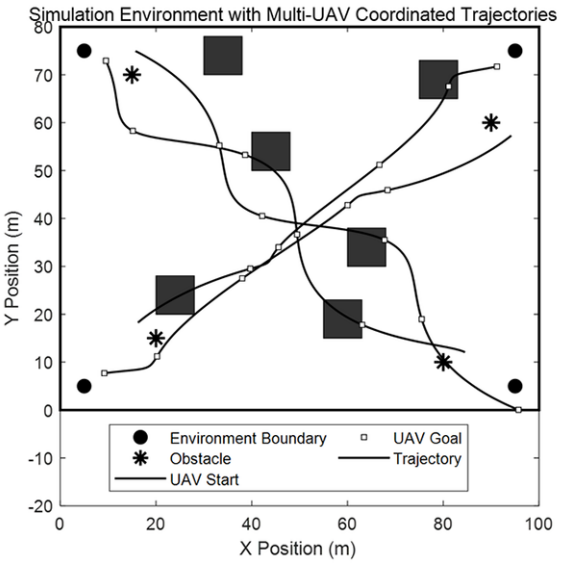
Dynamic Complexity: The integration of dynamic obstacles and network imperfections ensures the simulation reflects the unpredictability of real-world urban airspace.

Multi-UAV Coordination: The scenario supports collaborative operations among multiple UAVs, testing the model's ability to handle multi-agent interaction and task allocation.

Performance Metrics: The experiment focuses on metrics such as generalization error, computational complexity, response delay, and energy management efficiency, providing a multi-dimensional evaluation framework.

In summary, the design and setup of these experiments not only simulate the complexity of real-world applications but also lay a solid foundation for subsequent performance evaluation and analysis. The combination of MATLAB simulation and realistic scenario design ensures the validity and reliability of the experimental results, providing critical insights for the practical deployment of UAV trajectory optimization technologies (Hentati et al., 2023).

Figure 3. Schematic Diagram of Experimental Environment and UAV Trajectory Simulation



Experimental Result Analysis and Performance Evaluation

Based on the above experimental environment, this paper conducts a comprehensive and in-depth analysis and evaluation of the actual performance of the proposed deep learning trajectory optimization model. Key indicators such as trajectory optimization accuracy, real-time response performance, and network communication reliability were quantitatively measured in the experiments. The overall results show that the model proposed in this paper not only significantly outperforms traditional optimization methods but also clearly improves the shortcomings of existing deep learning models.

Specific analysis indicates that the model performs outstandingly in the accuracy and robustness of real-time trajectory adjustment, capable of quickly generating optimized responses when facing dynamic obstacles. Meanwhile, its performance in communication delay is also remarkable, significantly reducing network congestion and information delay through the adaptive communication mechanism. Additionally, in scenarios involving collaborative delivery tasks of multiple UAVs, the model achieves higher task completion efficiency with significantly more reasonable and efficient flight trajectories. Figure 4 demonstrates the differences in trajectory optimization performance between the proposed method and traditional methods, from which it can be clearly seen that the trajectory optimization strategy proposed in this paper leads comprehensively in key performance indicators.

Furthermore, to further illustrate the specific magnitude of performance improvements, this study designed a quantitative performance comparison table (Table 4). The data listed in the table further

confirm the excellent performance of the proposed method, such as significant reduction in response delay, effective control of generalization error, and superior energy consumption management. These experimental results fully demonstrate that the method proposed in this study not only has an advanced structure but also possesses high practical value in real-world application scenarios.

Figure 4. Curve Chart of Trajectory Optimization Performance Comparison Among Different Optimization Methods

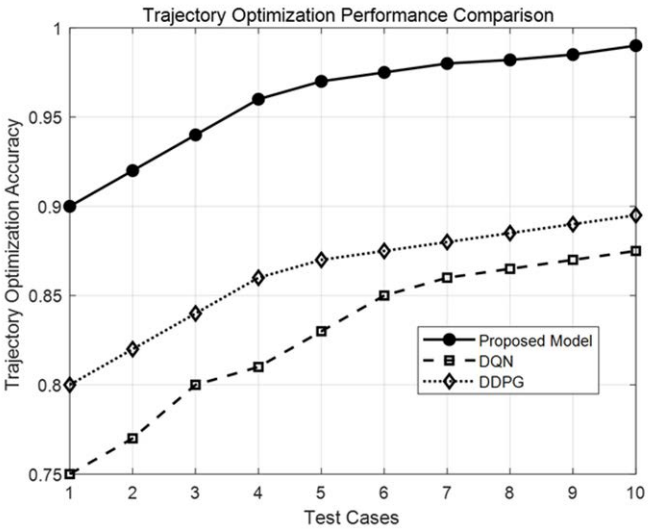


Table 4. Quantitative Performance Comparison Between the Proposed Method and Traditional Mainstream Methods

Performance Index	Proposed Model	DQN Model	DDPG Model
Response Delay (ms)	85	120	110
Generalization Error (%)	5.5	12.4	9.8
Energy Efficiency (%)	85	70	68
Communication Robustness (%)	92	78	80

DISCUSSION

This study proposes an optimization strategy integrating reinforcement learning and temporal deep neural networks to address the practical needs of trajectory optimization for networked UAVs. Experimental results demonstrate that the method excels in trajectory planning accuracy, real-time response capability, and communication efficiency, significantly outperforming traditional optimization approaches. However, it is important to acknowledge several limitations that warrant further improvement. For instance, while the simulation experiments strive to replicate complex real-world environments, they still have gaps compared to actual operational scenarios. Performance under extreme weather conditions or in the presence of complex urban obstacles requires more validation. Additionally, the trained model remains highly dependent on data, and its generalization performance may decline when environmental features differ significantly from the training data.

Although the adaptive communication control mechanism designed in this study significantly improves network delay issues, the dynamic allocation of communication resources has not been

thoroughly considered. For example, when the number of UAVs increases, resource competition in communication channels may emerge, potentially causing non-negligible impacts in practical applications. Therefore, future research could introduce more advanced communication resource optimization methods to further enhance the model's robustness and scalability. While this study has made progress in energy consumption management, the comprehensive consideration of UAV endurance remains insufficient. Future research should explore energy management potential from a more macro perspective, integrating multi-dimensional factors such as flight path optimization, power consumption patterns, and battery technology to develop a more comprehensive energy efficiency improvement strategy.

Furthermore, the current model primarily focuses on single-scenario trajectory optimization, while the adaptability to multi-task switching and dynamic mission reconfiguration needs enhancement. In urban delivery or emergency response scenarios, UAVs often need to adjust trajectories in real time based on changing tasks. Developing a more flexible task-driven model framework will be a critical direction for future research. Additionally, although the study validates the model's performance in multi-UAV collaboration, the coordination mechanism still lacks in-depth exploration of swarm intelligence. Introducing bio-inspired algorithms (e.g., ant colony optimization, particle swarm optimization) to improve collaborative efficiency and fault tolerance among UAV swarms could further expand the model's application scope.

In summary, while this research provides a novel approach for UAV trajectory optimization, its limitations in environmental adaptability, data dependency, communication resource management, and energy efficiency highlight the need for continuous improvement. Subsequent studies should focus on enhancing the model's generalization capability in extreme environments, developing data-efficient learning algorithms, optimizing communication-swarming coordination mechanisms, and constructing integrated energy management systems, thereby promoting the practical application and technological upgrading of networked UAVs.

CONCLUSION

This paper conducts a series of innovative research efforts focusing on the trajectory optimization problem of networked UAVs. First, it analyzes the numerous shortcomings of traditional trajectory optimization methods and existing deep learning models. Second, a deep learning model combining CNN and LSTM is proposed, innovatively introducing reinforcement learning and an adaptive communication mechanism. Finally, through simulation experiments, the performance of the proposed model is systematically verified, and a detailed comparative analysis is conducted with existing mainstream methods. The research results show that the method proposed in this paper has obvious advantages in multiple aspects such as trajectory accuracy, real-time response capability, communication delay control, and generalization performance. Especially in the ability to avoid real-time obstacles, this method demonstrates excellent robustness, significantly improving the operational safety of UAVs. Additionally, the communication robustness of the model makes UAV collaboration more stable and reliable, closer to real-world application requirements.

This study not only promotes the development of UAV trajectory optimization theory but also provides practical and effective technical paths and methodological guidance for its practical application, possessing strong practical significance. In the future, more in-depth research can be carried out on improving the model's generalization ability and optimizing communication resource management. Furthermore, simulations of more complex and variable real-world scenarios and field experiment verification are imperative. With the deepening of follow-up research, the optimization strategy proposed in this paper will have a broader application prospect in the field of networked UAVs.

DATA AVAILABILITY

The figures and tables used to support the findings of this study are included in the article.

CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

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Month 9, Year 2025
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AUTHOR CONTRIBUTIONS

Shouting Xin conceptualized and designed the research on networked UAV trajectory optimization, developed the deep learning model integrating CNN and LSTM with reinforcement learning, conducted simulation experiments, analyzed the results, and drafted the manuscript.

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